



Heaven's Light is Our Guide

Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology, Bangladesh

Course Code: CSE 2103
Course Title: Digital Logic Design

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Digital Techniques

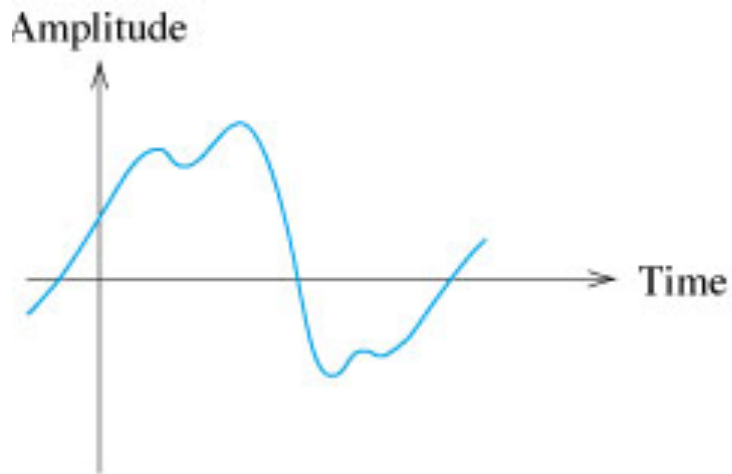
Technique

- technique of coming up with a solution to a problem

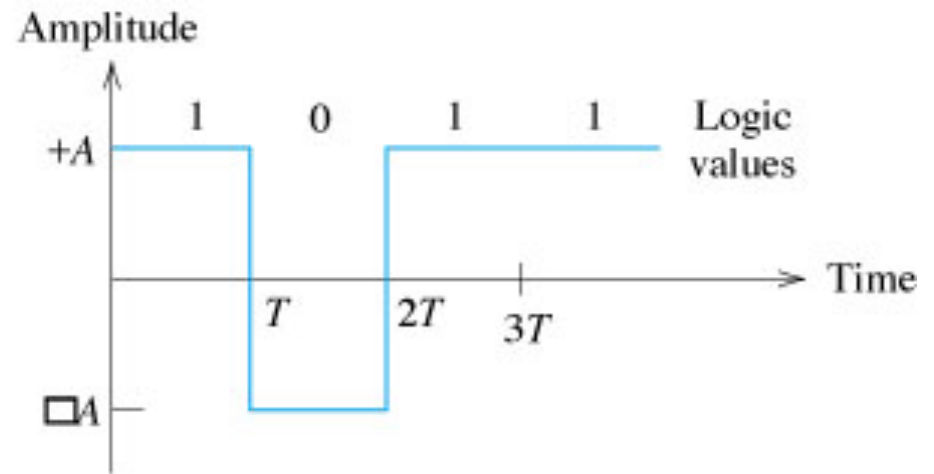
Digital Systems

- transform signals that can be abstracted as discrete in range and domain

Analog Vs Digital



(a) Analog signal



(b) Digital signal

Figure 1. Analog signals take a continuum of amplitude values.

Digital signals take a few discrete amplitudes.

Analog Vs Digital

Digital Circuits Advantages

- High noise immunity
- Easy IC implementation
- Highly adaptable
- Abstract-level design
- Cost-effective

Analog Circuits Advantages

- Fewer components
- Good for low-amplitude signals
- Excellent high-frequency performance

Analog Vs Digital System

- An **analog system** processes physical quantities that exist in a **continuous range of values**, where signals vary smoothly without distinct steps.
- A **digital system** consists of devices that handle **logical information or physical quantities in digital form**, where signals can assume only **discrete, separate values** rather than continuous ones.

Analog Vs Digital System

The cell phone has digital & analog components, and uses *both* types of signals.

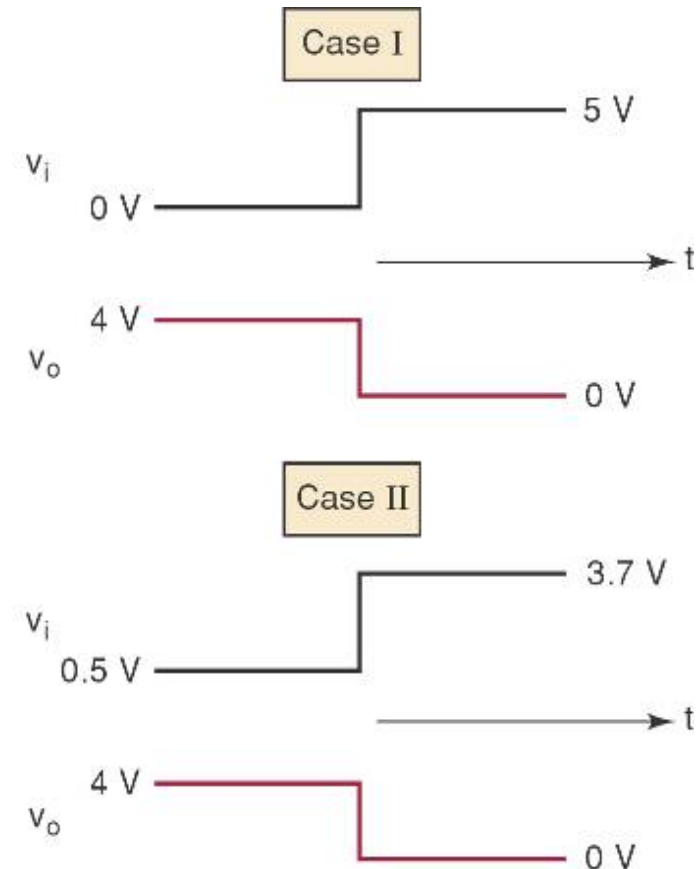


Digital Circuits/Logic Circuits

Digital circuits - produce & respond to predefined voltage ranges.

The term *logic circuits* is used interchangeably.

A digital circuit responds to an input's binary level of 0 or 1—not to its actual voltage.



Digital Computers

A **computer** is a hardware-based system capable of performing arithmetic operations, processing data, and making logical decisions.

It executes instructions from a program with **high speed** and **precision**, ensuring accurate and efficient task completion.

Digital Computers

Major parts of a computer:

Input unit—Processes instructions and data into the memory.

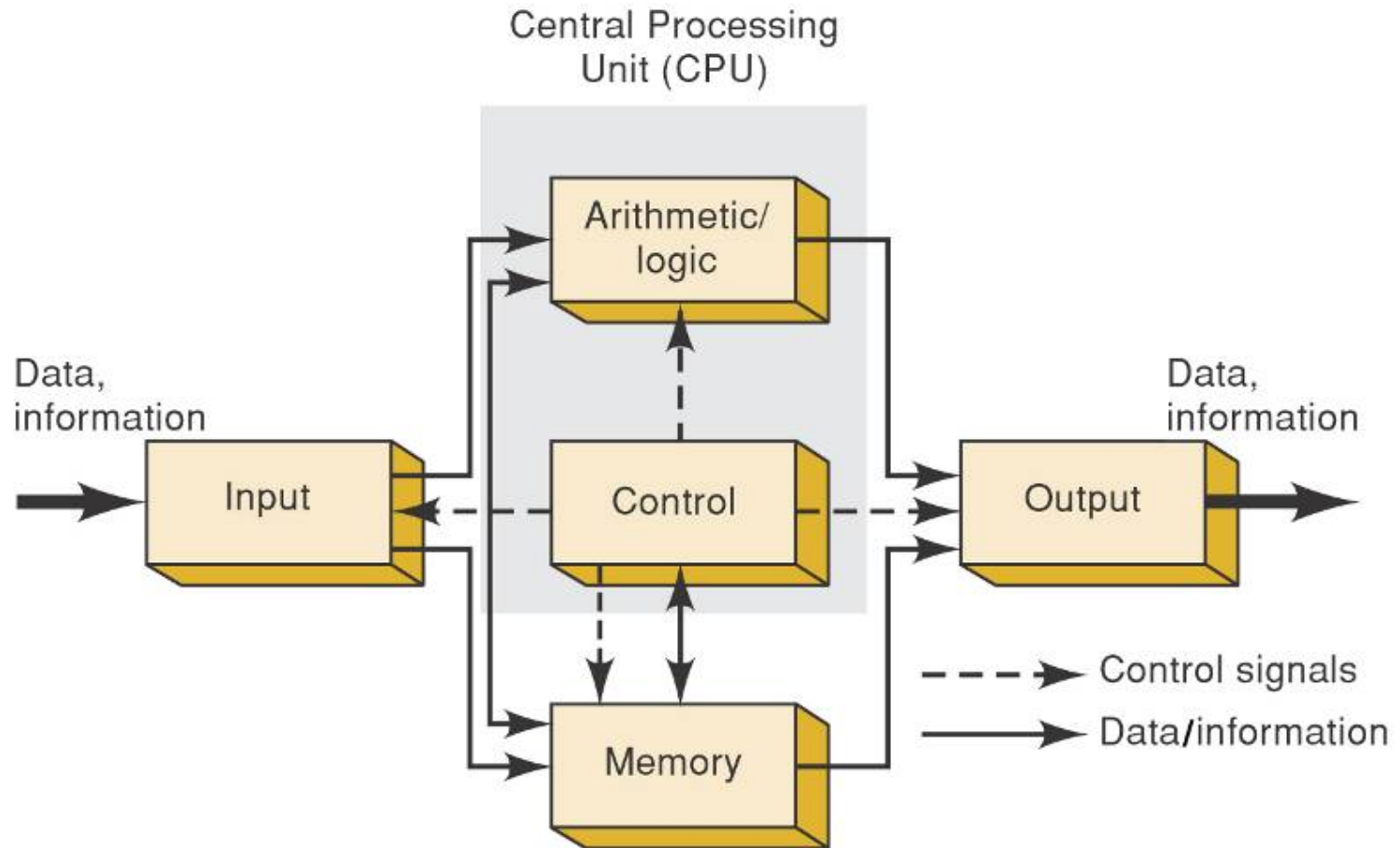
Memory unit—Stores data and instructions.

Control unit—Interprets instructions and sends appropriate signals to other units as instructed.

Arithmetic/logic unit—arithmetic calculations and logical decisions are performed.

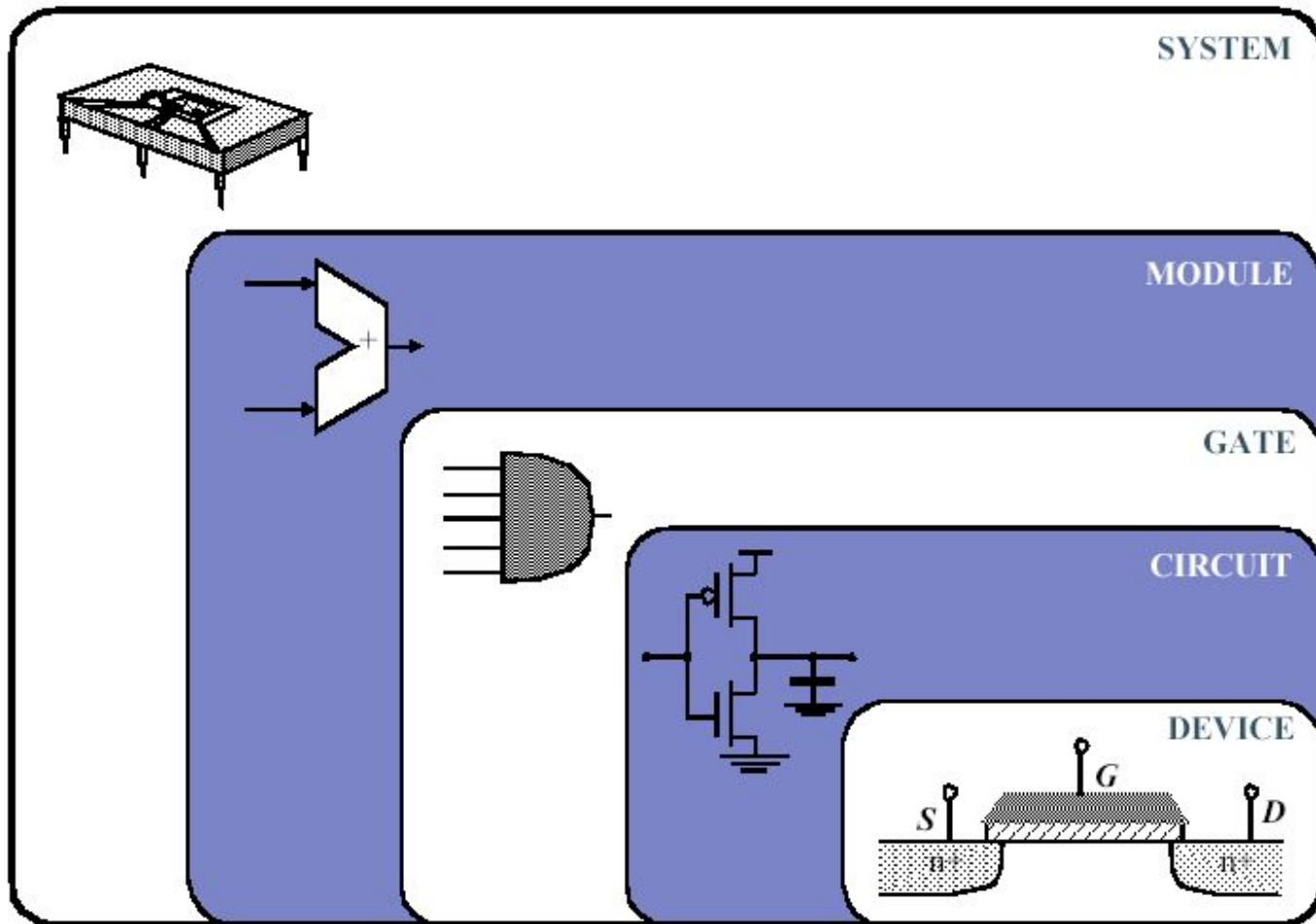
Output unit—presents information from the memory to the operator or process.

Digital Computers



The control and arithmetic/logic units are often treated as one and called the central processing unit (CPU).

Design Abstraction Levels



Digital Progress Today and Tomorrow

There are many needs in the world that digital technology can meet.

You will be able to become one of the pioneers on these new frontiers of technology.

Thank
You



Next topic